

# PROBLEM SCOPING

## USE CASE : AI BASED EARTHQUAKE RISK ASSESSMENT AND EARLY WARNING SYSTEM

### Disaster Management Department

- Notices that many areas suffer heavy damage due to earthquakes.
- Authorities want to reduce the impact of earthquakes and improve emergency response.

### They approach an AI team

- Asked to build an AI model/system to identify earthquake risk levels of different areas.
  - The system should help authorities prepare before an earthquake impact.
  - Sounds very straightforward.
  - But the solution statement is not a well-defined AI problem.
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### Solution:

### STEP 1. Understand the Business Problem

#### Initial Statement:

"Earthquakes are causing damage."

Need to uncover the actual problem.

The AI team will ask themselves multiple questions:

#### Questions for Stakeholders:

#### To Disaster Management Department

- What areas are most affected by earthquakes?
- How much damage occurs after an earthquake?

- How much time is available for emergency response?
- What resources are available during disaster management?

### **To Geological Department**

- What historical earthquake data is available?
- Which factors increase earthquake risk?
- How frequently do earthquakes occur in different regions?

### **To Government Authorities**

- Why is earthquake risk assessment important?
- What impact does earthquake damage make financially and socially?
- How can early warning help citizens?

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## **STEP 2. Define the Problem Properly**

### **Poor Problem Statement:**

"Predict when an earthquake will happen."

### **Better Approach:**

"Identify areas at high risk of earthquake damage using historical and real-time data so that authorities can prepare emergency responses before major damage occurs."

**What?** → Earthquake risk level prediction

**Who?** → Disaster management authorities and citizens

**When?** → Before earthquake impact

**Why?** → Enables better preparation and faster response

**Outcome?** → Reduce damage and improve safety

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## **STEP 3. Is AI Really Needed?**

Can the problem be solved without AI?

→ If only previous earthquake records are used, a simple rule-based system may work.

Example:

"If an area had previous earthquakes → mark it as high risk."

Why AI?

AI should only be used when:

- Patterns are complex.
- Manual rules are insufficient.
- Multiple factors are involved.

**In our case:**

Earthquake risk depends on multiple factors:

- Historical earthquake records
- Seismic activity
- Geological conditions
- Ground movement
- Population density

Patterns are complex.

**AI is needed.**

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## **STEP 4. Identifying Stakeholders**

**Stakeholders: Everyone affected by the project.**

**Stakeholder Mapping:**

**Disaster Management Department**

- Plan emergency response activities.

**Government Authorities**

- Create safety policies and disaster plans.

### **Geological Scientists**

- Analyze earthquake patterns and data.

### **Citizens**

- Receive early warnings and improve safety.

### **Rescue Teams**

- Get information for faster rescue operations.
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## **STEP 5. Define the AI Task**

### **Machine Learning Formulation:**

#### **Classification Problem**

Output:

- Low Risk
- Medium Risk
- High Risk

#### **Risk Scoring:**

- How likely is an area to be affected? (0-100%)

#### **Ranking Problem:**

- Which areas require immediate safety measures?

This aligns better with the disaster management objective.

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## **STEP 6. Define the Success Metrics**

This is critical.

**Technical Metrics:**

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

**Business Metrics:**

- Faster emergency response.
  - Reduction in damage.
  - Better disaster preparation.
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**STEP 7. Define Constraints**

Every AI project operates under constraints.

**Operational Constraint:**

- Emergency resources and rescue teams are limited.

**Time Constraint:**

- Risk analysis and warning should be generated quickly.

**Data Constraint:**

- Complete earthquake data may not exist for every region.
- Sensor data may be incomplete.

**Ethical Constraint:**

- Wrong predictions can create panic.
  - The system should provide reliable information.
  - The model should not be biased against any region.
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## **STEP 8. Assumption and Risk**

### **Assumptions:**

- Historical earthquake data represents future risk patterns.
- Sensors provide accurate information.
- Authorities act on AI recommendations.

### **Risks:**

- Earthquakes are difficult to predict accurately.
- Missing or incorrect data.
- False warnings may create panic.
- Unexpected environmental changes may affect results.